

Heard but Not Studied: Mapping the Gap Between Health Research and Women's Lived Experiences

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I. Abstract

Women represent just over half the global population, yet for decades they have been systematically underrepresented in biomedical research. Our project aims to help close this gap by assessing the most common health issues raised in women's online discussion forums and the extent of research published on these issues. This brings us to our research question, "What women's health symptoms and conditions are most discussed on r/WomensHealth, and to what extent are these topics represented in clinical research on PubMed and ClinicalTrials.gov?" To answer this, we performed topic modeling and keyword frequency counts of posts that anonymous users have submitted to Reddit's r/WomensHealth forum, obtained the most commonly discussed topics and their keywords, and used them to pull research data from PubMed and ClinicalTrials.gov, which are both hosted by the National Institutes of Health, and analyzed the gaps between discussion count, paper count, and trial count. We found that topics affecting both genders are 3.8 times more researched than those affecting only women. We also found that topics that had an expected paper count that was over 10 times the actual paper counts were probiotics (for vaginal health), Bartholin cysts, hormonal acne, unprotected sex, painful sex (Dyspareunia), interstitial cystitis, and bloating. These topics show the greatest disparity between discussion and research, prompting stakeholders to make data-driven decisions about which research to fund.

II. Introduction

Women's health is a crucial part of health research, directly addressing conditions that affect women. According to the World Health Organization, "Gender norms, power relations and structural inequalities influence women's exposure to health risks, access to care, quality of services and responsiveness of health systems. As a result, women may experience diseases differently from men, present with different symptoms and have distinct care needs at different stages of life (World Health Organization, n.d.)." Many factors contribute to women's health, but biological understanding and increased scientific research are critical to its development.

Historically, men's bodies were treated as the default in medical study design, with findings generalized to women despite important biological differences (Balch, 2024). Most of the time, this was not accidental; researchers often actively excluded female subjects due to concerns about hormonal variability interfering with results, the cost of housing female animals, and even due to policies designed to shield women of reproductive age from experimental drug exposure (Balch, 2024). The consequences have been serious, where women experience adverse drug reactions at twice the rate of men (Zucker & Prendergast, 2020), wait longer for both diagnosis and pain relief (Merone et al., 2022), and are more likely to be misdiagnosed during serious medical events (Merone et al., 2022). This is a pattern so well documented that it has been given a name, Yentl Syndrome, which is the phenomenon where a woman's illness must first mirror a man's presentation before it is taken seriously (Merone et al., 2022). Conditions

which affect women, including endometriosis, uterine fibroids, autoimmune disorders, and Polycystic Ovarian Syndrome (PCOS), remain underfunded and understudied relative to their prevalence (Edwards et al., 2023). Endometriosis is estimated to affect around 10% of women worldwide, yet diagnosis can take up to 10 years, in part because endometriosis is frequently dismissed as psychological (De Corte et al., 2024).

Online health communities have emerged as a large source of what burdens women experience in healthcare. Platforms like Reddit host active communities for conditions such as r/PCOS, r/endo, r/Fibrosis, r/periods, r/vaginismus, r/WomensHealth, and so much more. Women can share symptoms, treatments, healthcare experiences, and most importantly, gain knowledge. These discussions are extremely important, as they reflect women's lived experiences that are often missing from published research. Treating these spaces as data offers a way to locate that gap, or “missingness”, between what women experience versus what the research community prioritizes. To further investigate this, we have analyzed established research and clinical trials, specifically in PubMed and ClinicalTrials.gov, alongside what women frequently discuss on Reddit, to identify the most underrepresented areas of women's health research.

Research Question: What women’s health symptoms and conditions are most discussed on r/WomensHealth, and to what extent are these topics represented in clinical research on PubMed and [ClinicalTrials.gov](https://clinicaltrials.gov)?

III. Impact and Stakeholders

Women have historically been underrepresented and underprioritized within medical and clinical published research, which has shaped women’s ability to understand their own health issues, receive proper diagnoses, treatment, and easily navigate healthcare systems. If women continue to be left out of conversations in published health research, health systems will not stop burdening women with self-advocacy in spaces that have not been designed to support them. W. This project aims to deconstruct set precedents on women’s health by fully mapping what women discuss online, where published health research is absent, and which topics are most understudied, to establish the use of women's voices in future published research.

The primary stakeholder is women, whom this project directly impacts. Individuals who can directly use this project's findings in their work include, medical researchers, clinical trial designers, public health policymakers, and advocacy groups. As a result, medical researchers and clinical trial designers will have specific areas in women’s health that actively need to be researched and studied. Advocacy groups will better understand how to support women through research that includes women’s voices and uniting women to strengthen visibility. Patient advocacy organizations, for instance, the Endometriosis Foundation of America (Endometriosis Foundation of America, 2026) or the PCOS Awareness Association (*PCOS Awareness Association*, 2026), can use gap findings to strengthen funding arguments, and consequently shifting research priorities to better serve women. Public health policymakers can use this

project's findings, backed by data, to set regulations about health decisions directly related to women's needs.

Collectively, this project produces a data-driven, ranked map of women's health conditions that are widely discussed in patient communities but underrepresented in clinical literature, giving decision-makers a concrete, evidence-based starting point for prioritizing understudied conditions.

IV. Background & Related Work

Women face systemic barriers that influence who listens to, researches, and cares for them in our society. This is especially true in the health care system and in the biological understanding of women's bodies. Predominantly, research on the human body has used men as the baseline, as mentioned in section II, but legislative intervention has addressed this representation imbalance. In 1993, the U.S. Congress required the inclusion of women in federally funded clinical trials (Balch, 2024), and the 2016 Twenty-First Century Cures Act went further by mandating that trial results be stratified by sex, gender, race, and ethnicity (Balch, 2024).

Despite this progress, gaps remain substantial. Previous studies have evaluated National Institutes of Health (NIH) funding patterns. These studies have exposed the persistent disparities favoring diseases that predominantly affect men. Specifically, a study done by Kerri Smith on women's health funding found that conditions that affect women more are often less funded (Smith, 2023). It was found that conditions like migraines, headaches, endometriosis, and anxiety disorders, which all disproportionately affect women, receive less NIH funding in regard to the burden they cause. HIV/AIDS and substance misuse, which affect men more commonly, receive more funding (Smith, 2023). Many studies have made progress in quantifying the gap using participant data (Feldman et al., 2019). For example, a 2019 study quantified sex bias in clinical studies from ClinicalTrials.gov, using automated data extraction to assess the magnitude of female underrepresentation. They found substantial underrepresentation of women in 7 of the 11 disease categories they analyzed (Feldman et al., 2019). The most predominant areas were HIV/AIDs, chronic kidney diseases, and cardiovascular diseases. Categories of health that pertain just to women have even less research. For example, as of May 12, 2026, PCOS (Polycystic ovary syndrome) has officially been renamed to PMOS (Polyendocrine Metabolic Ovarian Syndrome) as a direct result of more research being done into women's health. The original name often misdirected understanding of this condition, leading to incorrect diagnoses by medical providers (Teede et al., 2026).

More recently, the World Economic Forum partnered with McKinsey Health Institute to create a Women's Health Impact Tracking platform to measure and close the women's health gap. This platform highlights and tracks statistics related to Clinical Trials published regarding women's health, mortality rates of conditions impacting women, and highlights indicator data for women's health (World Economic Forum, 2025). The conversation around women's health is

growing; more women are becoming aware of gaps in their knowledge about their own bodies. At the same time, social media and online communities are emerging, with Reddit as one of these spaces. A 2026 study by Weber Shandwick examined how women use Reddit as a place for community. They found discussion about women's health on the subreddit r/WomensHealth has increased by 91% in just a year. A key feature of Reddit is its anonymity, which creates a safe space for diverse experiences and perspectives. Intriguingly, women across generations use Reddit to share experiences and learn about their bodies, including perimenopause and age-specific health information (Weber, 2026).

New research on these kinds of spaces includes a 2025 study analyzing online communities formed around menopause (Funnell et al., 2025). Their main research angle was the quality of information in these communities, examined through a thematic analysis. They found many individuals use the internet as their main information source. However, little research compares women's lived experiences with published health literature. Our project aims to help close this gap.

V. Data & Data Engineering

Our project has three data sources: Reddit data obtained via Academic Torrents (Watchful1, 2026); and PubMed and ClinicalTrials data obtained using the Python packages PaperScraper (Born, 2026) and PyTrials (Cavalcante, 2024), respectively. Below, we outline the project's data extraction and engineering (Fig. 1). Our project is hosted on GitHub, where all our work is stored for reproducibility (Fig. 2).

Fig. 1. Full data ingestion and transformation pipeline

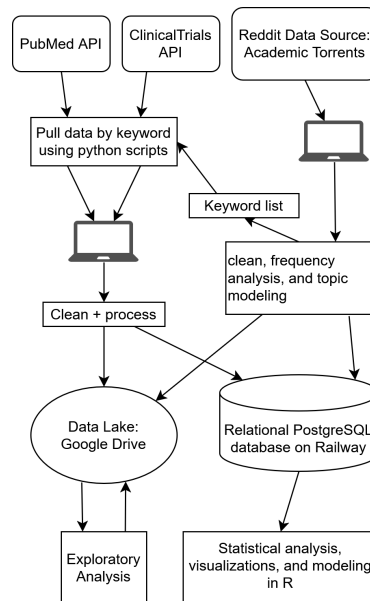
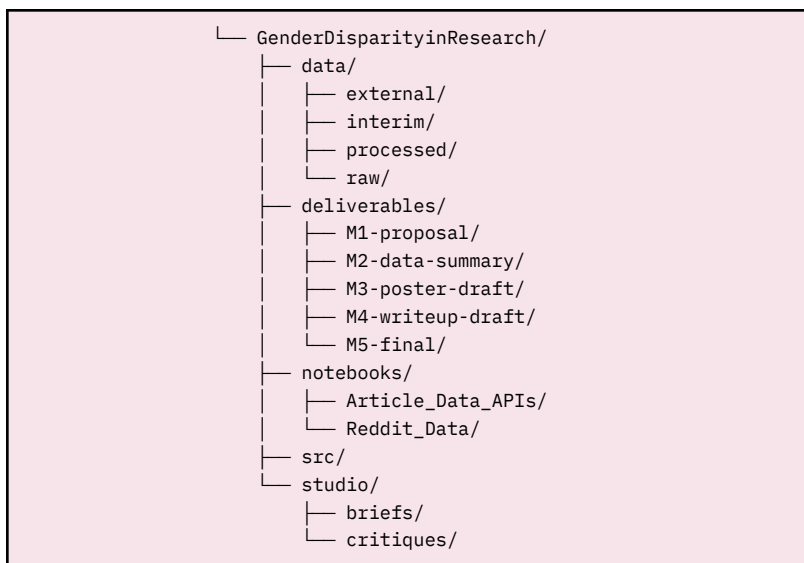


Fig. 2. GitHub repository structure



Reddit Data

The source contained all Reddit user posts and comments from 2005-06 to 2025-12 in the top 40k subreddits. We downloaded its torrent file, opened it with qBittorrent, selected the r/WomensHealth subreddit, and ran the torrent, which downloaded the subreddit's entire history of posts and comments. The raw data lives in **data/raw/subreddits25/** and has both **WomensHealth_comments.zst** (98MB) and **WomensHealth_submissions.zst** (43MB). We unpacked the submissions .zst file and put the contents in **data/interim/subreddits/WomensHealth/**, which is only a JSON file named **submissions** (370MB). We also uploaded all locally saved data to Google Drive, which served as our data lake for this project. We cleaned the data by removing null columns, null rows, and rows with deleted or non-English content. We conducted frequency analysis and used BERTopic to identify the most commonly discussed topics and compiled a list of keywords based on our results.

PubMed Data

We retrieved data from PubMed's API using the Python package PaperScraper, an open-source research software project that provides practical tools for scholarly metadata collection, publication search, full-text retrieval, citation analysis, and more. After obtaining the list of keywords, we wrote a Python script that iteratively queried for papers containing any of the keywords across different time blocks, spanning from 1786 to 2026. We saved these blocks as Pandas DataFrames, compiled them into a single DataFrame using `pd.concat`, saved it as a CSV in **data/raw/ as reddit_pubmed_data.csv**, and uploaded it to Google Drive. All of this work was done in **notebooks/reddit_keywords_pubmed_pull.ipynb**.

ClinicalTrials Data

The ClinicalTrials data was obtained in a similar fashion as the PubMed data. A Python package, PyTrials, was used to draw data using ClinicalTrials.gov's API. A script was created to

pull data across iterations of different time blocks and save each as a Pandas DataFrame. The ClinicalTrials data spans the years 1968 to 2026. We merged these separate DataFrames using `pd.concat` and saved the result as a CSV in `data/raw/` as `reddit_clinical_trials_data.csv`. All of this work was done in `notebooks/reddit_keywords_clinical_trials_pull.ipynb`.

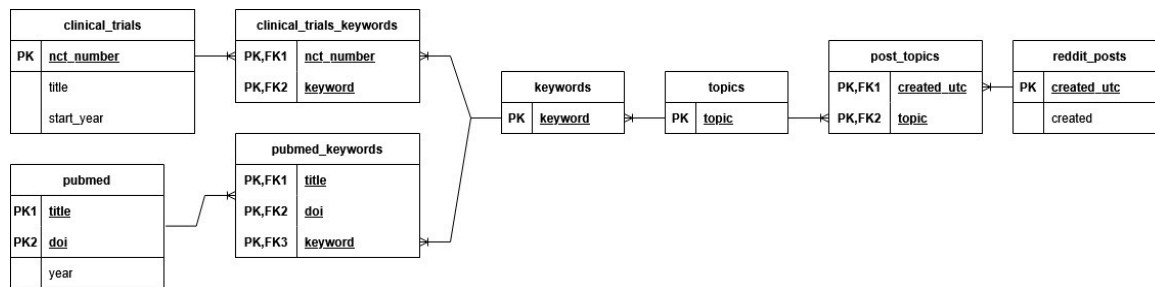
Google Drive

We used Google Drive as our data lake, a place to dump large amounts of data without incurring storage costs. It also serves as a place to collaborate on document files. In our Drive, we have four folders: **data/**, where we keep our large data files; **ERD/**, where we keep .drawio files of ERD sketches; and **notebooks/**, where we keep notebooks that utilize Google Colab's computing resources. Document files live in the home directory. All notebooks and anything necessary for reproducibility are copied to GitHub.

Relational Database

We have a PostgreSQL relational database hosted on Railway for a small monthly fee (Fig. 3). This is where we link together the relationships between data, keywords, and topics. This structure lets us query all clinical trials, PubMed papers, and Reddit posts linked to a topic. This database includes a view that aggregates the post, clinical trial, and paper counts per topic for easy export to a Pandas DataFrame for comparative analysis.

Fig. 3. Relational database for final dataframe



Final Dataset

The final dataset used in the analysis uses the three datasets above as the base. For each source, Reddit, PubMed, and Clinical Trials, each unique post, publication, or trial was counted for the final 49 topics. For example, the topic PCOS has 285 Clinical Trials, 6,916 PubMed publications, and 1,696 Reddit posts associated with it. This data, first a view, `topic_summary` was created in the PostgreSQL relationship database and downloaded as, `topic_summary.csv`, to use in Python and R for analysis.

VI. Ethics & Responsible Use

As this project works with health data, ethical considerations are imperative. The data needed for this analysis is extremely personal, and a Reddit post could potentially be linked to a person. This is why the username column is dropped entirely during data cleaning. Within the analysis, no personal information from the post title, comments, or post contents was used. Overall, this project acknowledges that Reddit data contains valuable experiential knowledge that is being used for a good cause. We acknowledge that using this type of data incorrectly can be misleading and divert attention from important topics.

PubMed publication data, as well as Clinical Trials data, were collected using Python scraper libraries with valid, open-source licenses. Data pulled from PubMed included study title, abstract, DOI, journal, authors, and year. No personal information was extracted from any columns pulled, and only raw counts were used in the final analysis dataframe. Data pulled from ClinicalTrials.gov included NCT Number, primary outcome measures, study title, sponsor, collaborators, sex, age, enrollment, summary, funder type, study type, start date, and completion date. No personal information was extracted from any columns pulled, and only raw counts were used in the final analysis dataframe.

This study is also limited by the data, which affects how the results can be interpreted. PubMed and ClinicalTrials.gov were the only publication data pulled. This does not capture research in other studies or from smaller publishers. Women's experience data was collected from Reddit, which has a very restricted audience: mostly individuals in the United States, those with access to technology, and those with knowledge of this community.

Finally, it is extremely important to note that this project is looking at disparities in women's health in research without taking into account the intersectionality of race and gender. Women of color face systemic components, including stereotyping, insurance access, and accessible healthcare, that all contribute to the imbalance of equity for racial groups within healthcare. Medical literature overall struggles to report data regarding sex and gender; it struggles even more so with reporting race and ethnicity (Bierer et al., 2022). As well as the intersection of race and gender, this project also does not take into account individuals outside of the gender binary who might still be affected by issues regarding the female reproductive system.

VII. Methods & Evaluation

Frequency Analysis

Frequency analysis is a computationally light way to identify the most commonly mentioned keywords, which is useful for assessing the topic makeup of r/WomensHealth at face value, assisting with data cleaning, and leading into more advanced topic modeling steps. We removed stopwords, common words that provide no insight into what the text is about ("and", "this", "of"), before creating frequency counts of keywords and bigrams (two-word keywords). To assess whether a count is statistically meaningful, a log-likelihood ratio/G-test can be done

instead of a Chi-squared test, as Chi-squared tests assume normal distribution when word counts are not normally distributed. For example, a study showed that words with a frequency of less than 1 in 50,000 make up about 20-30% of typical English news reports (Dunning, 1993). This means the distribution has longer tails than a normal distribution. Performing a G-test using log-likelihood is more accurate than a Chi-squared test, according to Dunning (1993). To do this on our dataset, we compute the frequency of our top keywords and compare it to their expected frequencies in English using the Python package wordfreq (Speer, 2022), to obtain a Bonferroni-corrected p-value using the G-test. The top 50 keywords had p-values near zero, verifying they occur more frequently in r/WomensHealth than in English.

BERTopic

BERTopic is an unsupervised topic modeling technique that groups documents by meaning rather than by shared vocabulary. It works in four stages. First, each post is converted into a sentence embedding, a numeric vector that places posts with similar meaning near each other even when they use different words, so that a post about "cramps so bad I can't stand up" falls near one about "debilitating period pain." Second, UMAP reduces those embeddings to a smaller number of dimensions so that clustering is computationally feasible. Third, HDBSCAN clusters the reduced embeddings. Unlike k-means, HDBSCAN does not require the number of topics to be set in advance, and it assigns posts that do not fit any dense cluster to an outlier group labeled topic -1 rather than forcing every post into a topic. Fourth, class-based TF-IDF (c-TF-IDF) treats all posts in a cluster as one document and ranks terms by how distinctive they are to that cluster, which produces the keyword list that describes each topic.

We applied BERTopic to the cleaned r/WomensHealth submissions. We joined post titles with post bodies, since titles carry a large amount of topical signal on Reddit, and removed URLs, placeholder text, and posts shorter than 15 characters. This left 68,848 posts. Topic keywords were generated with a CountVectorizer using English stopword removal, an n-gram range of one to two words so that clinically meaningful phrases such as "boric acid" and "night sweats" could appear as single terms, and a minimum document frequency of 5 to filter out rare tokens and misspellings.

The parameter that mattered most to our results was `min_topic_size`, which sets the smallest number of posts a cluster can contain. Our final model used a `min_topic_size` of 50 with `nr_topics` set to auto, producing 39 topics and assigning 39.7% of posts to the outlier group. We also tested a `min_topic_size` of 100, which reduced the model to 24 topics. That run lowered the outlier rate to 32.9%, but it did so by merging narrow clusters into two large and generic ones dominated by high-frequency words such as period, pain, and just. The clusters lost in that merge included boric acid use, vaginal probiotics, iron and ferritin levels, thyroid function, and pelvic floor therapy, which are the kinds of specific topics this project set out to find. Several of them later appeared among the most under-researched topics in our analysis. We kept the smaller minimum cluster size and accepted the higher outlier rate in exchange for that specificity.

Since our goal was to build a keyword vocabulary for querying PubMed and ClinicalTrials.gov rather than produce a topic model as a final product, we evaluated the output on interpretability and coverage rather than internal coherence scores. We manually reviewed the c-TF-IDF keywords and representative posts for each topic, and the resulting terms became the keyword list used for API retrieval, which is listed in Appendix A. A large outlier group is expected with this kind of data, since many posts on r/WomensHealth are broad requests for reassurance or long personal accounts that touch on several concerns at once and do not resolve into a single condition.

Topic Mapping

As each study from PubMed and ClinicalTrials was retrieved via their APIs, a keyword was assigned to each instance. However, if that keyword matched more than once to a study, that study or trial was repeated in the final datasets. This is also true for the Reddit data. To mitigate keyword repetition across studies, topic mapping was used to consolidate a single topic per study. We manually grouped each keyword by similar attributes and themes (see Appendix A). For example, all IUD keywords (intrauterine device, copper IUD, levonorgestrel IUD) were grouped under a single topic, IUD. After mapping keywords to topics, we ranked topics in a hierarchy that determined which one a record would be assigned when several applied. Specific diagnoses outrank general symptoms, so a study matching UTI, period cramps, and breast cancer is assigned to breast_cancer rather than to either symptom keyword. This produced 49 final topics, each with a PubMed publication count, a clinical trial count, and a Reddit post count. The complete keyword-to-topic mapping is listed in Appendix A, and the final topics are listed in Appendix B. One topic, boric_acid, returned a count of zero from ClinicalTrials.gov and a count of one from PubMed despite a substantial volume of discussion on r/WomensHealth. Because our ranking metric divides expected counts by actual counts, a topic with an actual count of zero produces an undefined ratio and a count of one produces an unstable one. This topic was therefore excluded. As mentioned in section V., Data & Data Engineering, the final analysis dataset, **topic_summary.csv**, contains the remaining 49 topics.

Model Testing

The final dataset used, topic_summary, and since the final analysis is count data with no values below zero, Poisson regression was used. The AIC (Akaike Information Criterion) baseline was set to gauge the performance of the models tested. This metric helps compare models by considering how well the model fits the data and how complex it is (ScienceInsights, 2026). A null model was fitted to both the PubMed publication count and the clinical trial count.

The first models used publications and trials based on post count; glm() was used in R, family = “poisson”. The AIC decreased for both models: for clinical trials, from 219700 to 214056, a decrease of 5644. PubMed decreased from 3486000 to 3215544, with a difference of 270,456. The post count coefficient was positive and highly statistically significant in both models. Based on this result, overdispersion was assessed to determine the next best steps for

modeling this data. PubMed, Clinical Trials, and Reddit post counts had extremely large variance relative to the mean. Poisson's distribution assumes the variance of the data to be equal to the mean, meaning the Poisson model needed to be changed to account for the overdispersion within the data. Quasi-Poisson was the next model tested. However, this model cannot produce an AIC value but does output a dispersion parameter for each model. This allows users to see how well overdispersion is being addressed. Once tested, the Quasi-Poisson model yielded a dispersion parameter of 8824.975 for clinical trials and 100929.1 for PubMed. This revealed that a different model was needed to handle overdispersion in the data. The next model tested was Negative Binomial Regression. Poisson only rescales the data's variance, not its shape. Negative Binomial Regression uses a different variance function in quadratic form to accommodate cases where the variance grows much faster than the mean (MetricGate, 2026). The AIC baseline for the null models was 812.9674 for clinical trials, and 1105.619 for PubMed. Negative Binomial Regression was tested for predictor, post count, and response, publication and trials. The AIC for clinical trials decreased from 812.9674 to 812.46, a difference of 0.5074. The AIC for PubMed decreased from 1105.619 to 1102.4, a difference of 3.219. Although the drop was extremely small, it indicates that Negative Binomial Regression is the best model for this data.

Table 1. Baseline AIC values for null models across distributions and outcome variables.

AIC	Model	Variable
219,700	Poisson	Clinical Trials
3,486,000	Poisson	PubMed
812.97	Negative Binomial	Clinical Trials
1,105.62	Negative Binomial	Pubmed

Note. All values are from intercept-only (null) models fitted to each outcome. Lower AIC indicates better fit relative to model complexity; values are comparable only within an outcome variable, not across PubMed and Clinical Trials. Quasi-Poisson models are omitted because that family does not produce an AIC.

Adding a New Variable

This project's final model uses a variable that was manually added and decided on based on further research. Due to data limitations, extracting sex or gender information from studies and trials was nearly impossible. Based on a topic and the keyword within that topic, each was assigned a boolean value if this keyword would be applicable to both genders or only women. For example, the "libido" topic, which contains keywords "libido" and "sexual desire", was assigned TRUE, as it pertains to both women and men. The topic "probiotics," containing the keyword "probiotics vaginal health," was assigned FALSE because it applies only to women. Topics like "breast cancer screening", which do affect both men and women but are overtaken by

research for women and affect women at a higher percentage, were assigned FALSE, female only.

VIII. Results & Visualizations

Exploratory Data Analysis

Before modeling, we examined how each of our three data sources is distributed across time. The three sources cover very different spans. Our Reddit data begins with the first post to r/WomensHealth in 2013 and runs through 2025. The ClinicalTrials.gov data spans 1968 to 2026, and the PubMed data spans 1786 to 2026. All three show the same underlying shape: slow accumulation followed by sharp, roughly exponential growth in recent decades or years.

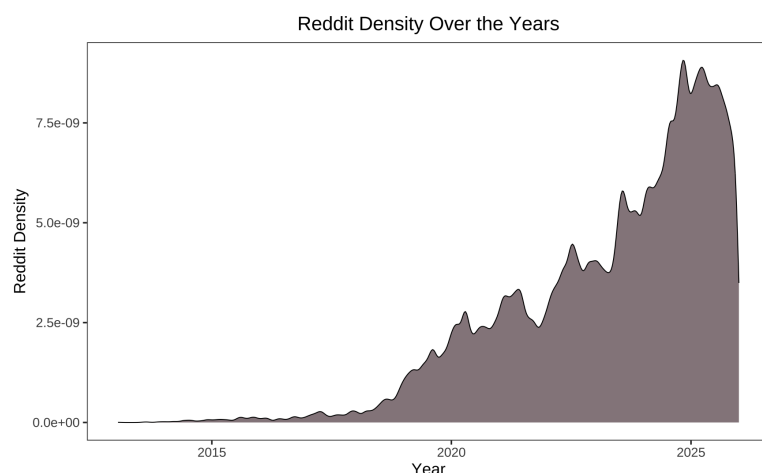


Fig. 4. Reddit post density over time (2013 - 2025)

Reddit post volume grows steadily from the subreddit's founding in 2013, accelerates sharply after 2019, and continues climbing through the end of the collection window. Part of this growth reflects the general expansion of Reddit as a platform rather than any change in women's health

concerns specifically, but the pattern confirms that the community is active and growing rather than dormant, and that the bulk of our discussion data reflects recent experience.

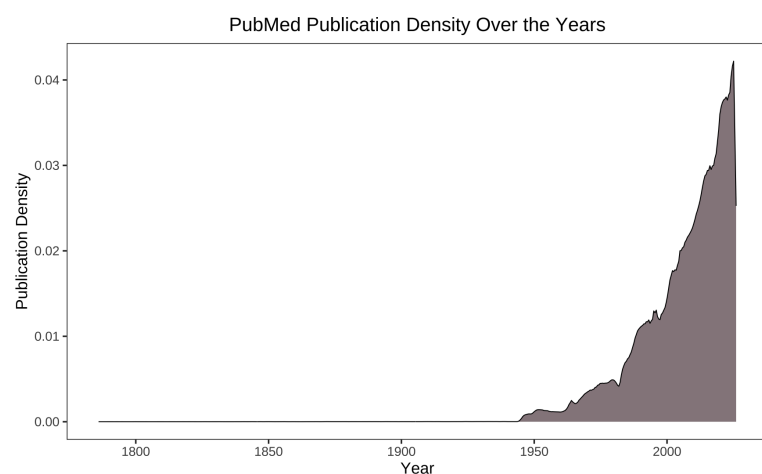


Fig. 5. PubMed publication density over time (1786 - 2026)

PubMed publication counts are effectively flat until the mid-twentieth century, then rise steeply from roughly 1950 onward, with the sharpest increase occurring after 2000. Although the corpus technically reaches back to the eighteenth century, the overwhelming majority of

matching publications were published within the last few decades.

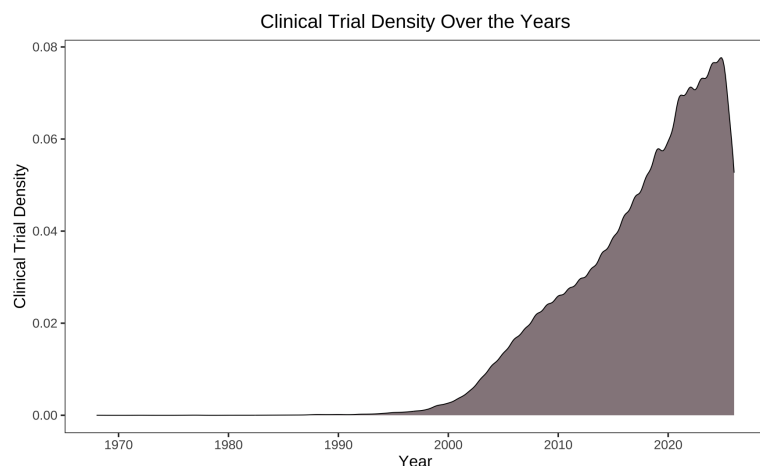


Fig. 6. Clinical trials density over time (1968 - 2026)

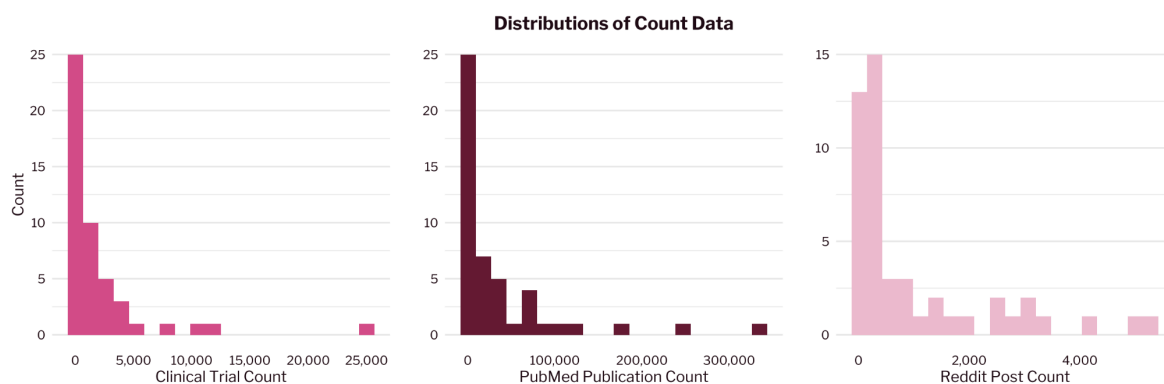
Registered clinical trials show a similar but more compressed pattern. Volume is negligible before 2000, which is when the ClinicalTrials.gov registry was established, and grows rapidly thereafter. Trials with earlier start dates appear in the data because they were registered retrospectively.

The difference in time span across our three sources reflects the research question rather than a flaw in the data. We are not asking whether research responded to online discussion or whether discussion followed research. We are asking how much medical knowledge exists on the topics women are discussing now.

Framed that way, the full historical record of publication is correct, because research does not expire. A study of Bartholin cyst treatment published in 1970 is still available to a clinician treating a patient today. Restricting the literature to a window matching our Reddit data would understate the body of knowledge a woman or her physician can actually draw on, which is the quantity we want to measure.

This choice is also defensible statistically. Including the full publication history inflates paper and trial counts, but by a roughly constant factor across all topics, and a constant multiplier of that kind is absorbed into the model intercept. It does not affect the coefficient on sex applicability or the expected-to-actual ratios we use to rank topics, since both depend on how counts vary between topics rather than on the absolute scale of any one source.

Fig. 7. Final dataset count distribution for each source



We also examined how each count variable is distributed across the 49 topics in our analysis dataset. All three are heavily right-skewed. Most topics cluster near zero while a small number reach counts orders of magnitude larger, with clinical trials extending past 25,000, PubMed publications past 300,000, and Reddit posts past 4,000. In every case, the variance far exceeds the mean, a pattern known as overdispersion. This matters for model selection, since Poisson regression is the standard approach for count data but assumes that the variance of the response equals its mean. Fig. 7 shows that assumption does not hold for any of our three variables, which is why the modeling process described below moves from Poisson to negative binomial regression.

Fig. 8. Most frequent topics from each source, color coded by sex affect.

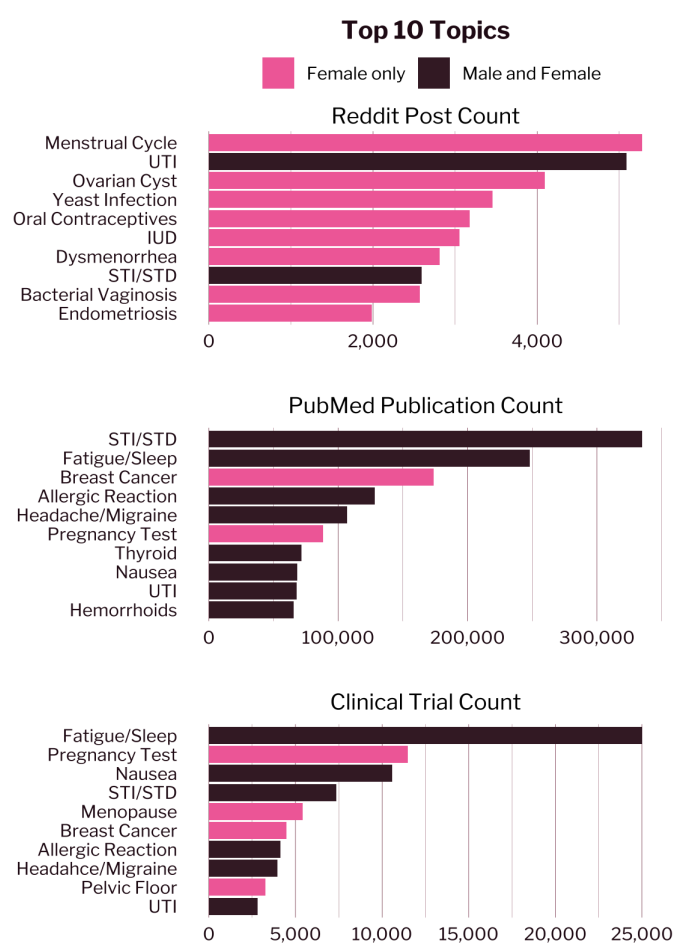


Fig. 8 shows the ten most frequent topics in each of our three sources, color coded by whether the topic affects women only or both sexes. The contrast between discussion and research is immediate. Among the ten most discussed topics on r/WomensHealth, eight affect women only, including menstrual cycle, ovarian cysts, yeast infections, oral contraceptives, IUDs, dysmenorrhea, bacterial vaginosis, and endometriosis. Among the ten most published topics on PubMed, only two do, and among the ten most studied topics on ClinicalTrials.gov, only four do.

The topics that dominate the research lists are largely general conditions that happen to appear in a women's health context: fatigue and sleep, allergic reaction, headache and migraine, nausea, and hemorrhoids. These rank near the top of

both publication and trial counts while appearing nowhere in the top ten of what women actually discuss. Conversely, the specifically gynecological topics that lead the Reddit rankings are almost entirely absent from the top of the research rankings, with UTI and STI/STD the main exceptions, both of which affect both sexes. This reversal previews the central result of our analysis. Topics affecting both sexes accumulate research at a much higher rate than topics affecting only women, and that gap holds even before accounting for how much each topic is

discussed. The models in the following section quantify the disparity while controlling for discussion volume and impact on sex type.

Predicted Counts Modeling

We ran two negative binomial regression models predicting paper counts and trial counts from post counts. We log-transformed the post count variable to account for a heavily skewed count distribution. We organized topics into two groups: True if the topic affects both sexes; False if the topic affects only females. We added that as a predictor variable in our model.

PubMed Paper Counts

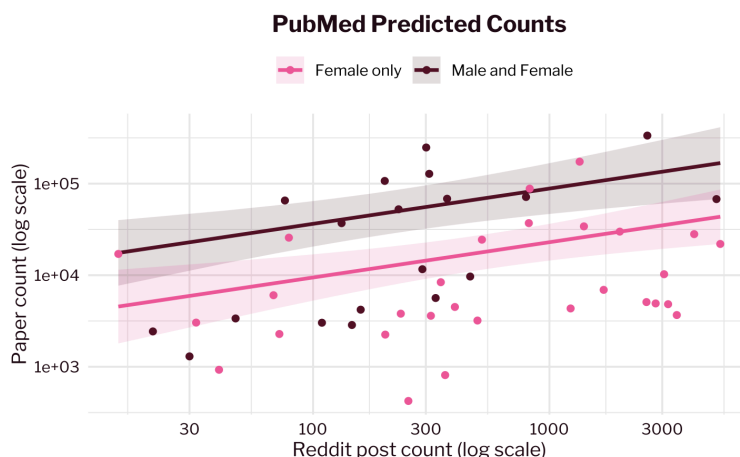
Table 2. Negative binomial regression predicting PubMed paper count from Reddit post count and sex-specificity of topic ($n = 49$ topics).

Predictor	Estimate	Std. Error	z	p	$\exp(\beta)$
(Intercept)	7.3792	0.7662	9.631	< .001 ***	—
log(post count)	0.3852	0.1169	3.296	< .001 ***	1.47
both_sexes = TRUE	1.3493	0.3677	3.669	< .001 ***	3.85

Note. Model: **glm.nb(paper_count ~ log(post_count) + both_sexes)**. Dispersion parameter for the Negative Binomial(0.6949) family taken to be 1. $\theta = 0.695$ ($SE = 0.121$). Null deviance = 79.668 (47 *df*); residual deviance = 58.072 (45 *df*); AIC = 1078.3; $2 \times \log\text{-likelihood} = -1070.33$. Significance codes: *** $p < .001$, ** $p < .01$, * $p < .05$.

The coefficient for post count is 0.3852, indicating that a 1% increase in posts is associated with a 0.3852% increase in papers. However, the coefficient for both_sexesTRUE tells us something entirely new: topics that affect both sexes have about 3.8 ($\exp(1.3493)$) times as many papers as topics that affect only females. Fig. 9 shows the best-fit line for each sex category, where we can visually see the disparity.

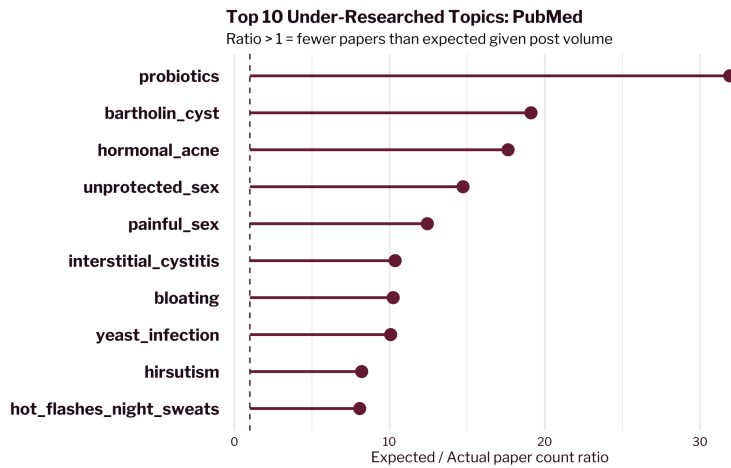
Fig. 9. Regression plot for prediction of PubMed counts based on Reddit post counts.



Using the model to estimate an expected paper count based on both_genders and post_count, we can compare the actual paper count to the expected count by calculating ratios. Fig. 10 shows the top under-researched topics with the

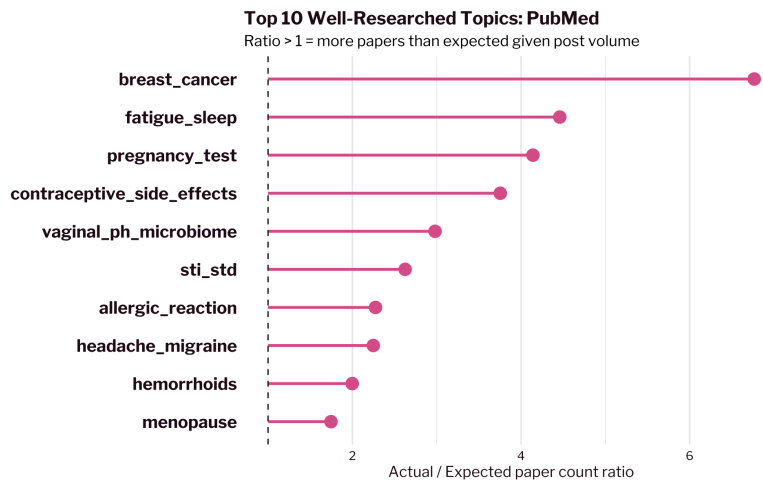
largest expected-to-actual paper-count ratios.

Fig. 10. The most under-researched topics based on expected counts from the model for PubMed publications.



The most under-researched topic is vaginal health probiotics, with an expected-to-actual paper count ratio of 31.8, and Bartholin Cyst the second most under-researched topic, with a ratio of 19.0. Fig. 11 shows the relatively over-researched topics, with the largest actual-over-expected ratio.

Fig. 11. The most well-researched topics based on expected counts from model for PubMed publications.



The most-researched topic is breast cancer, with an actual-to-expected ratio of 6.7.

ClinicalTrials.gov Count

Table 3. Negative binomial regression predicting ClinicalTrials.gov trial count from Reddit post count and sex-specificity of topic ($n = 40$ topics).

Predictor	Estimate	Std. Error	z	p	exp(β)
(Intercept)	4.3809	0.8748	5.008	< .001 ***	—

log(post count)	0.4187	0.1334	3.138	.002 **	1.52
both_sexes = TRUE	1.3278	0.4198	3.163	.002 **	3.77

Note. Model: `glm.nb(clinical_trial_count ~ log(post_count) + both_sexes)`. Dispersion parameter for the Negative Binomial(0.5335) family taken to be 1. $\theta = 0.534$ ($SE = 0.091$). Null deviance = 73.777 (47 *df*); residual deviance = 60.200 (45 *df*); AIC = 797.29; $2 \times \log\text{-likelihood} = -789.29$. Significance codes: *** $p < .001$, ** $p < .01$, * $p < .05$.

The coefficient for post count is 0.4187, showing that a 1% increase in posts leads to a 0.4187% increase in trials. However, the coefficient for both_sexesTRUE tells us something entirely new: topics that affect both sexes have about 3.8 ($\exp(1.3278)$) times as many trials as topics that affect only females. Fig. 12 shows the best-fit line for each sex category, where we can visually see the disparity.

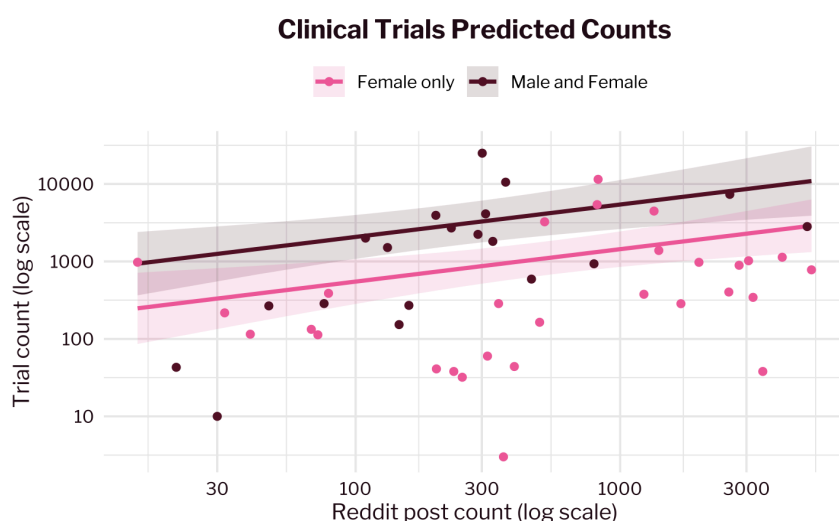
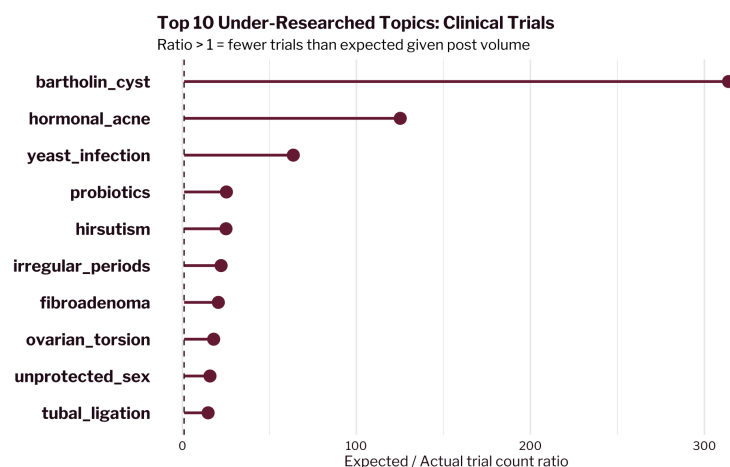


Fig. 12. Regression plot for prediction of Clinical Trial count based on Reddit post count.

Using the model to estimate an expected trial count based on both_genders and post_count, we can compare the actual trial count to the expected count by calculating ratios. Fig. 13 shows the top

under-researched topics with the largest expected-to-actual trial-count ratios.

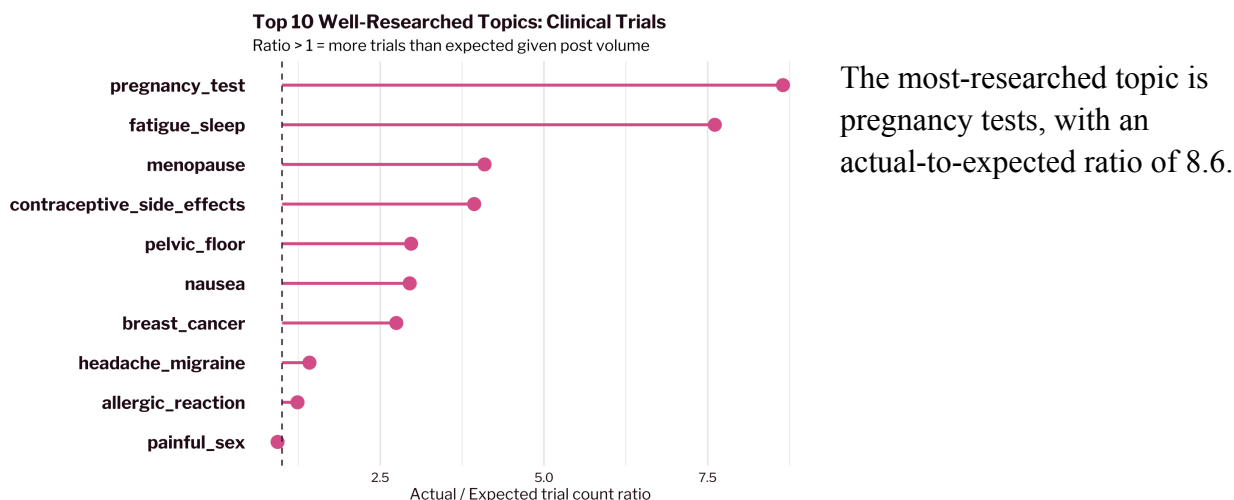
Fig. 13. The most under-researched topics based on expected counts from the model for Clinical Trials.



The most under-researched topic is Bartholin Cyst, with an expected-to-actual trial count ratio of 313.9, and hormonal acne is the second most under-researched topic, with a ratio of 125.2. Fig. 14 shows the relatively over-researched

topics, with the largest actual-over-expected ratio.

Fig. 14. The most well-researched topics based on expected counts from model for Clinical Trials.



IX. Discussion, Limitations, & Future Work

Discussion

This project identified key topics women discuss, yet publications and trials remain very few. Across ClinicalTrials, the top five most underrepresented topics include: Bartholin cyst, hormonal acne, yeast infections, vaginal probiotics, and hirsutism (See definitions in Appendix C). Throughout PubMed, the top five most underrepresented topics include: vaginal probiotics, Bartholin cyst, hormonal acne, unprotected sex, and painful sex. Between PubMed and Clinical Trials, many underserved topics overlapped, highlighting the significance of these topics and the lack of research. Alongside identifying areas needing support, we also found the most well-researched topics relevant to the discussion. For Clinical Trials, this includes: Pregnancy test, fatigue/sleep, menopause, contraceptive side effects, and pelvic floor. For PubMed: Breast cancer, fatigue/sleep, pregnancy test, contraceptive side effects, and vaginal pH microbiome. Similarly, the well-researched topics in PubMed and ClinicalTrials.gov are relatively similar, showing consistency across locations and types of health research.

In addition, the analysis showed that topics affecting both sexes were researched 3.8 times more often, confirming previous research on the gap in women's health research. Fig. 8 maps the frequency of papers, trials, and posts by topic, showing how this relates to the statistical finding. A clear difference emerges between Reddit's top topics and those of PubMed and ClinicalTrials.gov: topics that mainly affect women appear much less often in research. These results are consistent with prior work approaching the gap from different angles. Feldman et al. (2019) found women substantially underrepresented as participants in 7 of 11 disease categories,

and Smith (2023) found that conditions disproportionately affecting women, including migraine, endometriosis, and anxiety disorders, receive less NIH funding relative to the burden they cause. Our findings extend this pattern to a third dimension, showing that the disparity appears not only in who is enrolled and what is funded, but in the sheer volume of published literature available on the conditions women discuss most.

Limitations

As stated previously in VI. Ethics & Responsible Use: this project uses only two online publication websites and one online community (Reddit). The limitation is as follows: these results cannot be generalized to the full global scope of women's health research and to all women's experiences. The purpose of this study was to gain a deeper understanding of prior research on gender disparity in the medical field and to initiate a discussion in which women's voices are heard. Another limitation concerns how keywords are used. Both PubMed and ClinicalTrials.gov data sources were retrieved using keywords based on Reddit users; this alone is a limitation, as Reddit Users will not always use the vocabulary used in published research. As studies were pulled, keywords were collected if that word appeared in any text on the front API page. For example, if a Clinical Trial is about heart disease but includes postmenopausal women, or menopause is a key variable, then "menopause" would be pulled as a keyword for that study. Some manual keyword checks were conducted to evaluate the pull's accuracy, but due to the dataset size, a full accuracy check was not possible. Finally, referring back to VII. Methods & Evaluation, keyword mapping is another limitation of this project. Naturally, a study or trial will relate to many different words, and most of the time it is more complex than a single topic. The hierarchical mapping was implemented to mitigate this limitation, but with the available data, an accurate capture was not attainable.

Future Work

Future work is necessary to advance this area. Stakeholders will be able to use the results to explore why women discuss these issues more than others, prioritize these topics in their research, and use data-driven findings to guide funding decisions. Foremost, this project should be replicated using more data. This means including more research publications and trials from more journals, small and large, as well as obtaining more discussion data. Survey data should be utilized to gain a more nuanced understanding of what women talk about, what impacts them the most, and what they struggle with. Using survey data will also help to include individuals who are not often captured in online data. Methods of analysis should also be expanded by using more in-depth text analysis to include more context from studies. For example, key concepts, gender, or patient count can be extracted from the abstract or body text of research papers using Named Entity Recognition or Semantic Embeddings. Ultimately, future work should build on this project's results and goals.

X. Conclusions & Recommendations

Earlier, we demonstrated how we calculated the ratio of expected-to-actual paper counts for each topic. After topic modeling of the Reddit forum r/WomensHealth, topics with significant discussion were selected to be included in our analysis, so they all have a real impact on women's lives. The topics with PubMed paper expected-over-actual ratios of more than 10 include: probiotics (for vaginal health), Bartholin cysts, hormonal acne, unprotected sex, painful sex (Dyspareunia), interstitial cystitis, and bloating. Topics with expected-over-actual clinical trial ratios of more than 50 include: Bartholin cysts, hormonal acne, and yeast infections.

These topics show the greatest disparity between discussion and research, prompting stakeholders to make data-driven decisions about which research to fund. Our research also showed that topics appearing in r/WomensHealth that impact both sexes receive about 3.8 times the research as topics affecting only the female sex. This project has begun incorporating women's voices into medical and clinical discussions that have historically excluded them.

XI. Reproducibility & References

Reproducibility

All code is hosted on GitHub here:

<https://github.com/Addie-Shanti-Org/GenderDisparityinResearch>

A requirements.txt file is set up to ensure the proper environment settings. Our raw data is hosted on Kaggle here:

Clinical Trials:

<https://www.kaggle.com/datasets/shanti33/womens-health-clinical-trials-dataset-1968-2026/data>

PubMed: <https://www.kaggle.com/datasets/shanti33/womens-health-pubmed-dataset-1786-2026>

Reddit:

<https://www.kaggle.com/datasets/addieg/reddit-postscomments-rwomenshealth-as-of-dec-25/data>

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XII. Appendix

Appendix A. Keyword to Topic Mapping

Table A1. Search keywords used to query PubMed and ClinicalTrials.gov, grouped by assigned topic.

Topic	Keywords
menstrual_cycle	menstrual cycle
irregular_periods	irregular menstruation
heavy_bleeding	menorrhagia, heavy menstrual bleeding
abnormal_bleeding	abnormal uterine bleeding, breakthrough bleeding, intermenstrual bleeding
dysmenorrhea	dysmenorrhea, period cramps
amenorrhea	amenorrhea
oral_contraceptives	combined oral contraceptives, progestin-only pill
iud	IUD, intrauterine device, copper IUD, levonorgestrel IUD
contraceptive_side_effects	contraceptive side effects, birth control and mood, birth control and libido
tubal_ligation	tubal ligation
yeast_infection	vulvovaginal candidiasis, recurrent yeast infection
bacterial_vaginosis	bacterial vaginosis
uti	urinary tract infection, UTI
boric_acid	boric acid vaginal suppository
vaginal_ph_microbiome	vaginal pH, vaginal microbiome
probiotics	probiotics vaginal health
pelvic_pain	pelvic pain
ovarian_cyst	ovarian cyst

ovarian_torsion	ovarian torsion
endometriosis	endometriosis
adenomyosis	adenomyosis
pelvic_floor	pelvic floor dysfunction, pelvic floor physical therapy
vulvodynia	vulvodynia
interstitial_cystitis	interstitial cystitis
bartholin_cyst	Bartholin cyst
breast_lump	breast lump
fibroadenoma	fibroadenoma
breast_pain	breast pain, mastalgia, nipple pain
breast_cancer_screening	breast cancer screening, mammography, breast ultrasound
pcos	PCOS, polycystic ovary syndrome
hormonal_acne	hormonal acne
hirsutism	hirsutism
thyroid	thyroid dysfunction, hypothyroidism, Hashimoto's thyroiditis
menopause	perimenopause, menopause hormone therapy, HRT
hot_flashes_night_sweats	hot flashes, night sweats
sti_std	sexually transmitted infection, STI
painful_sex	painful intercourse, dyspareunia
libido	libido, sexual desire
unprotected_sex	unprotected sex
pregnancy_test	pregnancy test
abortion	medical abortion, misoprostol, mifepristone
abortion_policy	abortion access

hair_loss	hair loss, alopecia
iron_anemia	iron deficiency anemia, ferritin
bloating	bloating
hemorrhoids	hemorrhoids
nausea	nausea
fatigue_sleep	fatigue, sleep disturbance
heart_palpitations	heart palpitations
headache_migraine	headache, migraine
allergic_reaction	allergic

Appendix B. Final Topics and Sex Applicability

Table B1. Final analysis topics, the topics merged into each, and whether the topic was coded as applicable to both sexes.

Final Topic	Topics Merged	Applies to Both Sexes
bartholin_cyst	bartholin_cyst	No
ovarian_torsion	ovarian_torsion	No
ovarian_cyst	ovarian_cyst	No
endometriosis	endometriosis	No
adenomyosis	adenomyosis	No
fibroadenoma	fibroadenoma	No
breast_cancer	breast_lump, breast_cancer_screening	No
interstitial_cystitis	interstitial_cystitis	Yes
tubal_ligation	tubal_ligation	No
abortion_policy	abortion_policy	No
abortion	abortion	No

uti	uti	Yes
bacterial_vaginosis	bacterial_vaginosis	No
yeast_infection	yeast_infection	No
sti_std	sti_std	Yes
pcos	pcos	No
thyroid	thyroid	Yes
menopause	menopause	No
vulvodynia	vulvodynia	No
iud	iud	No
oral_contraceptives	oral_contraceptives	No
contraceptive_side_effects	contraceptive_side_effects	No
pregnancy_test	pregnancy_test	No
boric_acid	boric_acid	No
probiotics	probiotics	No
vaginal_ph_microbiome	vaginal_ph_microbiome	No
dysmenorrhea	dysmenorrhea	No
amenorrhea	amenorrhea	No
heavy_bleeding	heavy_bleeding	No
abnormal_bleeding	abnormal_bleeding	No
irregular_periods	irregular_periods	No
unprotected_sex	unprotected_sex	Yes
pelvic_floor	pelvic_pain, pelvic_floor	No
painful_sex	painful_sex	Yes
breast_pain	breast_pain	No

libido	libido	Yes
hirsutism	hirsutism	No
hair_loss	hair_loss	Yes
iron_anemia	iron_anemia	Yes
menstrual_cycle	menstrual_cycle	No
hormonal_acne	hormonal_acne	Yes
bloating	bloating	Yes
nausea	nausea	Yes
fatigue_sleep	fatigue_sleep	Yes
headache_migraine	headache_migraine	Yes
allergic_reaction	allergic_reaction	Yes
heart_palpitations	heart_palpitations	Yes
hemorrhoids	hemorrhoids	Yes
hot_flashes_night_sweats	hot_flashes_night_sweats	Yes

Note. Applies to Both Sexes indicates whether the topic was coded as affecting both sexes (Yes) or only women (No). The boric_acid topic is included here but was excluded from the regression models because it returned one paper and zero trials.

Appendix C. Definitions

Abnormal uterine bleeding. Bleeding between monthly periods, prolonged bleeding, or an unusually heavy period.

Adenomyosis. A condition in which the tissue lining the uterus grows into the uterine wall, causing the uterus to enlarge.

Amenorrhea. The absence of menstruation, commonly defined as missing one or more periods or as not having begun menstruating by age 15.

Bacterial vaginosis. A common vaginal infection caused by the overgrowth of bacteria normally present in the vagina, resulting in a bacterial imbalance.

Bartholin cyst. Also called a vulvar cyst, a fluid-filled cyst that forms near the vaginal opening on either side of the labia. It develops at the Bartholin glands, which produce lubricating fluid.

Boric acid vaginal suppository. A medication used to help restore vaginal acidity and to relieve symptoms of yeast infection such as itching and burning.

Breakthrough or intermenstrual bleeding. Bleeding or spotting that occurs between normal menstrual periods or during pregnancy.

Dysmenorrhea. Recurrent painful periods with no identifiable underlying cause.

Dyspareunia. Painful sexual intercourse.

Endometriosis. A condition in which tissue similar to the uterine lining grows outside the uterus.

Fibroadenoma. A common non-cancerous breast lump that occurs in people who menstruate.

Hashimoto's thyroiditis. Also called chronic lymphocytic thyroiditis, an autoimmune disease in which the thyroid gland is gradually destroyed.

Hirsutism. Excessive body hair growth in areas where hair is normally minimal or absent.

Hormone replacement therapy (HRT). Treatment that supplements declining levels of estrogen, progesterone, or testosterone.

Interstitial cystitis. A chronic condition causing bladder pressure, bladder pain, and sometimes pelvic pain.

Intrauterine device (IUD). A small T-shaped device inserted into the uterus to prevent pregnancy.

Levonorgestrel IUD. An intrauterine device that releases progestin to prevent pregnancy.

Mammography. Breast imaging using low-dose x-rays to detect cancer before symptoms appear, when it is most treatable.

Mastalgia. Breast pain.

Menopause. The point at which menstrual cycles stop, typically occurring between ages 45 and 55, accompanied by hormonal changes.

Menorrhagia. Heavy menstrual bleeding, defined as losing more blood than is typical during a menstrual cycle.

Ovarian cyst. A small sac filled with fluid or semisolid material that forms on or within one or both ovaries.

Ovarian torsion. A condition in which an ovary twists on the tissues that support it.

Pelvic floor dysfunction. A common condition in which a person cannot properly relax and coordinate the pelvic floor muscles used for urination and bowel movements.

Perimenopause. The transition to menopause, during which hormone levels decline and symptoms such as hot flashes and irregular periods begin. It can last several years.

Polycystic ovary syndrome (PCOS). Recently renamed polyendocrine metabolic ovarian syndrome. Diagnosed when at least two of the following are present: irregular menstrual periods, high testosterone or related symptoms, or an excess of antral ovarian follicles on ultrasound.

Progestin-only pill (POP). A form of birth control containing only progestin, a medication that mimics progesterone to inhibit ovulation and thicken cervical mucus.

Tubal ligation. A surgical sterilization procedure in which the fallopian tubes are permanently blocked, clipped, or removed.

Urinary tract infection (UTI). An infection affecting any part of the urinary system.

Vaginal microbiome. The community of microorganisms, including bacteria, viruses, and fungi, that live in the vagina.

Vaginal pH. A typical vaginal pH falls between 3.8 and 4.5. This slightly acidic level supports beneficial bacteria and protects against infection. Infection, hormonal changes, and sex can all shift vaginal pH.

Vulvodynia. A chronic pain condition affecting the vulvar area that occurs without an identifiable cause.

Vulvovaginal candidiasis. An overgrowth of yeast within the vagina, most often *Candida albicans*.